

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester III Examination, 2022-23

Paper No. CHB-301: Organic & Physical Chemistry-II

Time: 3 Hours

Maximum Marks: 70

Instructions: Answer five questions in total, including Question 1 from each section which is compulsory. Use separate answer books for Section A and Section B.

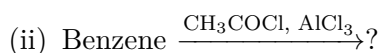
SECTION A

(Organic Chemistry-II – Marks: 35)

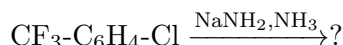
Question 1. Answer all parts of the following:

[5 × 2 = 10]

- (a) Classify the following species as aromatic, antiaromatic, or nonaromatic: (i) Cyclopropenyl cation, (ii) Cyclopentadienyl anion, (iii) Cycloheptatriene, and (iv) Cyclooctatetraene.
- (b) Among the following, which has the most acidic α -hydrogen? Write the appropriate reason: (i) Acetone, (ii) Acetylacetone, (iii) Ethyl acetoacetate, and (iv) Acetaldehyde.
- (c) Complete the following reactions, predicting the major product:



- (d) Write the main product in the following reaction, showing the mechanism:



- (e) Write the correct product of the following electrophilic aromatic substitution reaction with an appropriate reason:

**Question 2.**

- (a) Draw the molecular orbital energy level diagram of benzene. How would you explain that all carbon-carbon bond lengths in benzene are identical? [4]
- (b) Compare the relative reactivity of nitrobenzene and phenol towards electrophilic aromatic substitution reactions. [4]
- (c) Write the mechanisms of the following reactions: [4]
- (i) Mannich reaction
- (ii) Acid-catalyzed aldol condensation

Question 3.

- (a) What is positive and negative hyperconjugation? Explain with suitable examples. [4]
- (b) Write the mechanisms of the following rearrangement reactions: [4]
- (i) Hofmann rearrangement
 - (ii) Haloform reaction
- (c) Describe the DDT synthesis reaction mechanism and discuss its environmental uses and impacts. [4]
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Question 4.

- (a) Discuss the Reimer-Tiemann reaction with a detailed stepwise mechanism. [4]
- (b) Compare and explain the relative reactivity of acid chloride, acid anhydride, amide, and ester with a nucleophile. [4]
- (c) Explain the mechanism for the synthesis of diazonium salts. How would you obtain the following from benzenediazonium chloride? [4]
- (i) Thiophenol
 - (ii) Phenyl isocyanide
 - (iii) Biphenyl
 - (iv) Phenylhydrazine
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SECTION B

(Physical Chemistry-II – Marks: 35)

Question 5. Answer all parts of the following: [5 × 2 = 10]

- (a) What is liquid junction potential? How can it be minimized?
- (b) Compare the transference numbers of the chloride ion (Cl^-) in HCl and NaCl solutions.
- (c) The resistance of a 0.5 M solution of an electrolyte in a cell was found to be 45 ohm. Calculate the molar conductance of the solution if the electrodes of the cell are 2.2 cm apart and have an area of 3.8 cm^2 .
- (d) Calculate the Miller indices of the crystal planes for which the intercepts along the crystallographic axes are $(a/2, 2b/3, \infty c)$.
- (e) Write the Gibbs phase rule and define the terms involved taking some suitable examples.

Question 6.

- (a) How is the molar conductivity at infinite dilution (Λ_m^∞) determined for acetic acid (CH_3COOH) and hydrochloric acid (HCl)? Explain using Kohlrausch's law. [6]
- (b) What would be the pH of a solution in a cell if it generates 0.833 V e.m.f. at 25°C , when it is connected to a saturated calomel electrode (SCE)? [4]
- (c) What do you mean by a reversible electrode? Discuss any one reversible electrode mentioning its Nernst equation for electrode potential. [4]

Question 7.

- (a) Discuss the determination of transport numbers using the moving boundary method. [6]
- (b) What is the degree of freedom for a system consisting of a saturated aqueous solution of NaCl in contact with solid NaCl and its vapours? [4]
- (c) Draw and discuss the phase diagram of the lead-silver (Pb-Ag) system. What is the eutectic point? [4]

Question 8.

- (a) What is the Nernst distribution law? Describe the procedure to determine the equilibrium constant of the reaction $\text{KI} + \text{I}_2 \rightleftharpoons \text{KI}_3$ in aqueous medium using the Nernst distribution law. [6]
- (b) Define space lattice and unit cell of a crystal. [2]

- (c) Write a short note on different types of point defects (Schottky and Frenkel defects) in solids. [3]
- (d) Derive Bragg's equation for X-ray diffraction. What is the significance of n in this equation? [3]